

Modeling and Simulation of Mineral Processing Plants

An Introduction to the Use of Plant Simulation

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Module 1

This module is designed to introduce you to the graphics editor in MODSIM. The editor is used to draw and edit flowsheets and it is necessary to become fluent with the editor in order to use MODSIM effectively.

If you have never used MODSIM before you should look at the first few viewlets in the MODSIM TRAINER

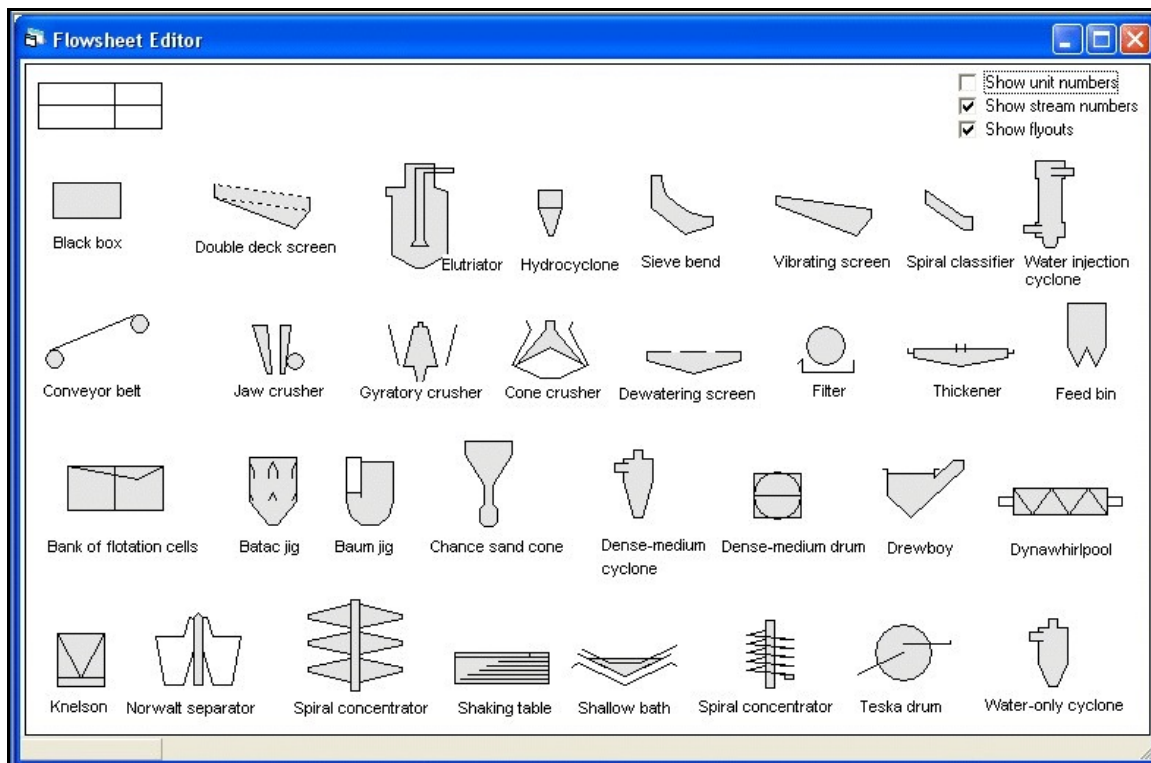
The exercises are grouped into sections for easy reference.

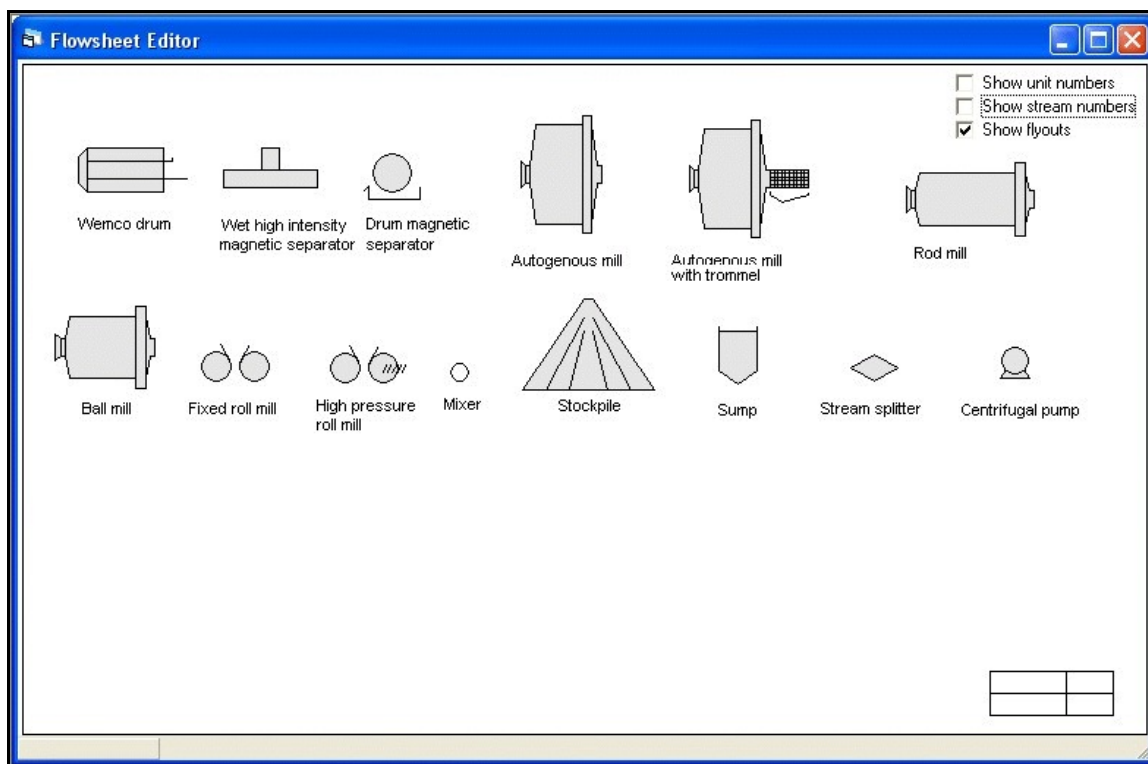
Drawing operations

These exercises will give you some practice with the drawing operations that are used in the graphics editor.

Exercise 1-1 Drawing the unit icons.

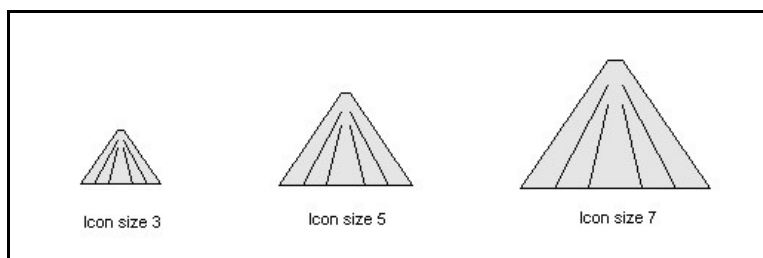
Open a new job in MODSIM and draw one copy of each of the unit icons. You will not be able to fit them all on to a single sheet. When you have filled a sheet, close the job without saving it and start a new job. Before closing the flowsheet, add an annotation to each icon and finally use Move on the Edit menu to rearrange icons and annotations to make a neat drawing like those shown below.





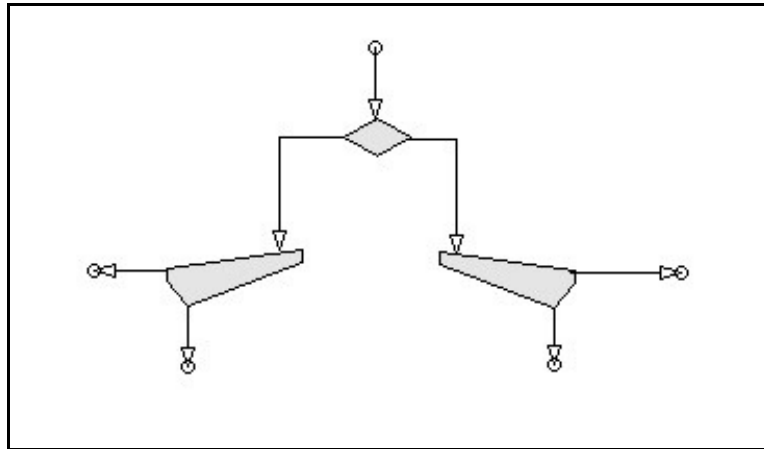
Exercise 1-2 Changing icon size.

Start a new flowsheet and draw three stockpiles of different size.



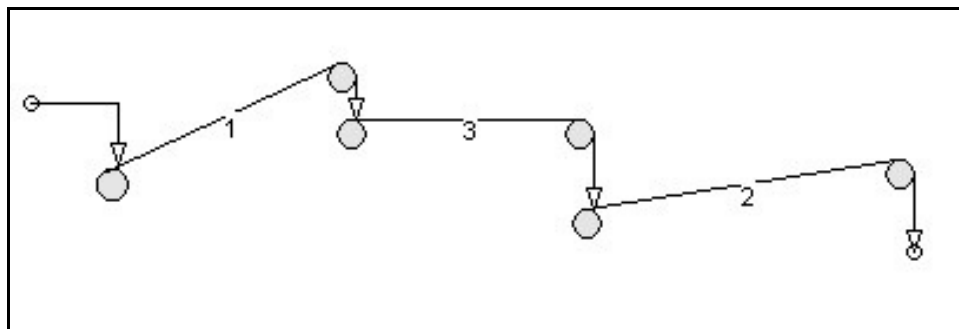
Exercise 1-3 Reflecting icons.

Draw two vibrating screens on a new flowsheet with one in the reflected orientation.



Exercise 1-4 Drawing conveyor belts.

Draw several conveyor belts of different lengths and in different orientations.



Editing operations.

Exercise 1-5 Editing flowsheets.

Start MODSIM and retrieve the Exercise 1-5 job from the Internet.

Edit the flowsheet and transform it as shown below.

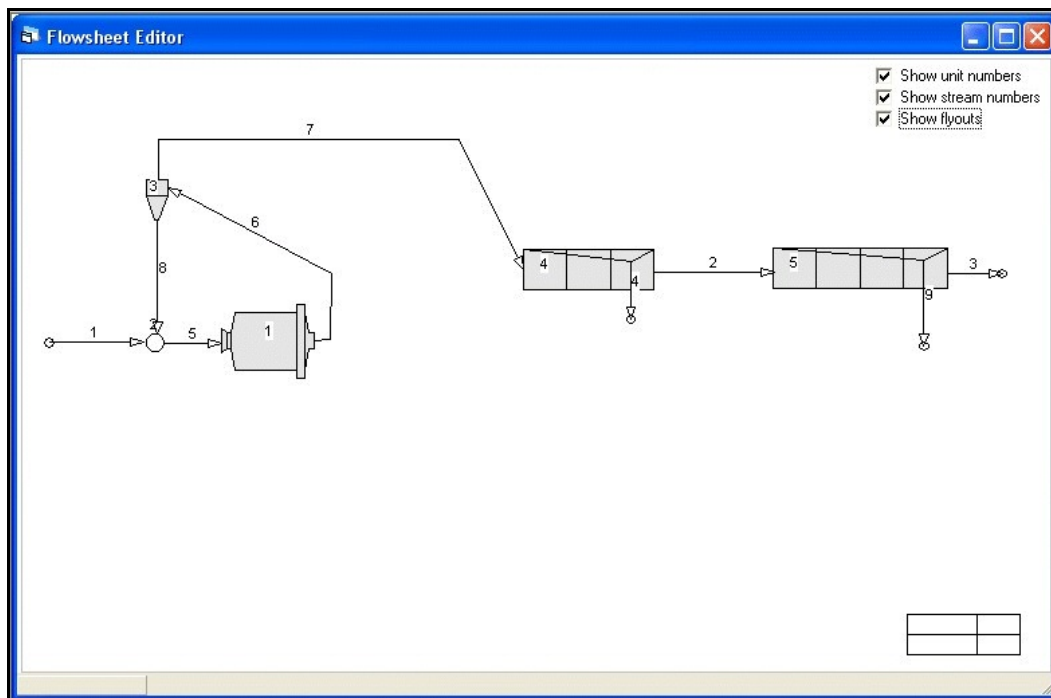
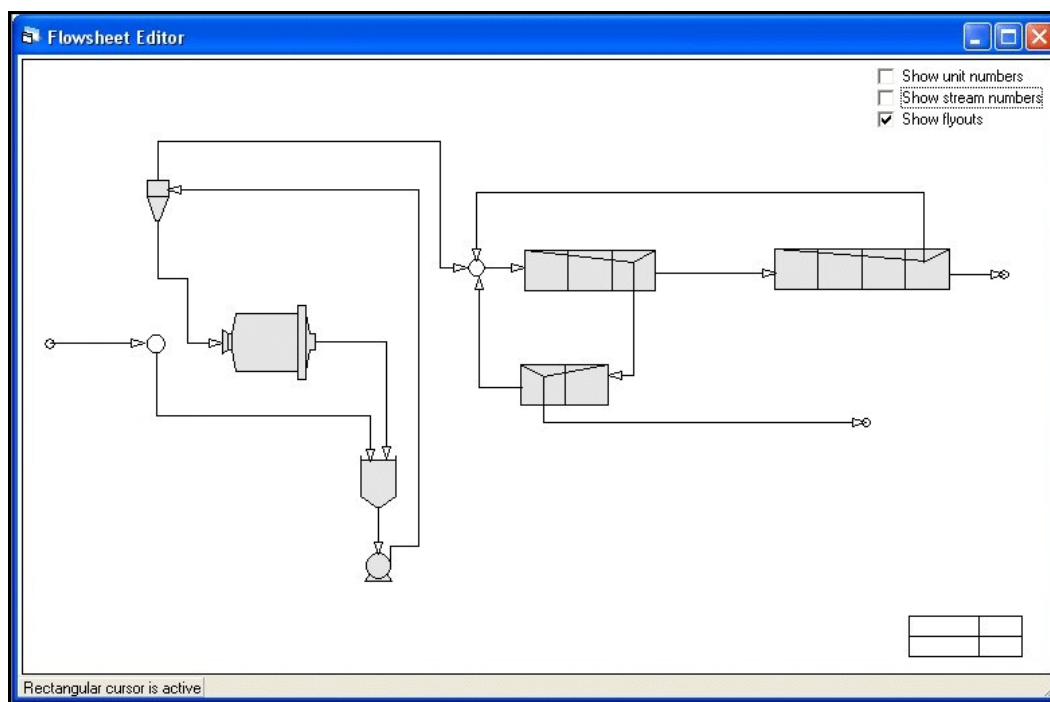


Figure 6 An untidy flowsheet



You can read a short description of simulation in Modeling and simulation of Mineral Processing Plants, Chapter 10. A short presentation is also available by running the program Simulation.exe

Module 2

Data specification.

Before the operation of a plant can be simulated it is necessary to specify precisely the characteristics of the material that is to be processed and the parameters that define the behavior of the individual unit operations in the flowsheet.

MODSIM uses the population balance method to model the operation of the units. This method requires the particulate nature of the solids to be specified using distribution functions. The particle size distribution is familiar and this is the basic distribution function that is used. However it is usually necessary to describe other properties of the particles as well as the size and provision is made for this as well. The basic principles of distribution functions and population balance methods are described in Chapter 2 of Modeling and Simulation of Mineral Processing Systems

MODSIM makes a clear distinction between data that is needed to describe the ore and the data that is necessary to define the behavior of the unit models.

The objectives of this module are

1. Learn to specify the properties of the ore - the system data.
2. Learn to specify parameters for the unit models.
3. Learn to find and analyze the data produced by the simulator.

System data

The system data includes any information that applies to the flowsheet as a whole. This includes information that MODSIM needs to set up its calculation algorithms such as for example the number of size classes that it must use for all calculations, the number of minerals that it must keep track of and the liberation characteristics of the minerals in the ore. In some cases additional physical properties must be specified as well. The following exercises will illustrate the procedures for specifying these data.

Parameters for the unit models.

Each unit model will require some parameters that define the operating conditions for the unit. These need to be specified before the plant is simulated. The models that are available in MODSIM are very varied so that the data required for each unit model is unique. Furthermore the parameters required for the different units that use the same model may very well be different. It is therefore necessary to specify model parameters for every unit in the flowsheet.

Exercise 2-1: Basic system data and the particle size distribution of the feed.

The specification of the data will be illustrated using a simple exercise based on simulating the performance of a single hydrocyclone.

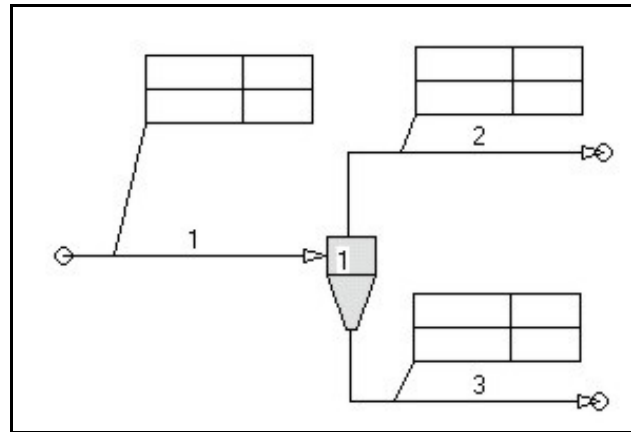
Start a new job in MODSIM and draw a flowsheet containing only a single hydrocyclone. Add a stream for the feed and one each for the overflow and the underflow. Add flyouts to each of the streams. Accept the flowsheet, save the job and start to edit the system data.

The material to be processed will contain only 1 mineral - silica of specific gravity 2.7. Enter this information on the form. Choose to specify specific gravity by mineral.

The hydrocyclone is primarily a size classification device and obviously the size distribution in the feed material will be of particular importance and must be specified before the operation of the hydrocyclone can be simulated. Other important properties of the feed will be the solids content and the feed rate.

The density of the solid material will also influence behavior of the particles in most units and this must be specified before any simulation is attempted.

The basis of the population balance method is the calculation of the behavior of each type of particle in the equipment that is being simulated. To do this, the particle population is conceptually divided into many classes in such a way that the particles in a single class are all similar to each other. The behavior of each class of material is then calculated and the overall behavior is calculated by accumulating the results from each particle class. In general the calculations produce more accurate results the more classes that are used and the narrower the class intervals. But of course the amount of calculational effort will increase as the number of classes increases. MODSIM allows particle populations to be distributed in up to 3 dimensions. The first dimension describes the particle size. The second dimension describes the mineral composition or grade of the particles and the third dimension is available to describe any other particle property that might be useful in any particular simulation. It is necessary therefore to consider carefully what level of detail should be used to describe the solid material so that the results of the simulation are useful and meaningful. In this exercise we will simulate the classification of a homogeneous solid so that



ORE CHARACTERISTICS

Select type of mineral to be processed

☒ Conventional minerals

☐ Coal processing plant

Number of minerals Mineral specific gravities

Mineral names

Specify sp.gr by

☐ Mineral

☒ Particle type

Number of size classes Largest particle size m

Number of grade classes Number of S-classes

only the particle size is relevant. The solid phase should be described by as many size classes as possible to enable MODSIM to model the hydrocyclone in sufficient detail. The student version of MODSIM allows a maximum of 25 size classes so choose this value for this exercise. Because no other properties of the solid are relevant in this exercise, no further distribution of the particle population need be considered. Thus the number of grade classes will be 1 and the number of S-classes will also be 1. MODSIM also requires a specification of the largest particle size that will be relevant in the problem. MODSIM uses this size to set up the size classes that it will use internally to do the modeling and simulation calculations. The classes are constructed using a square root of 2 geometrical sequence starting at the largest size and working downward for the specified number of size classes. In this exercise a largest size of 2mm will be satisfactory. Please note that this size must be specified in meters (ie 0.002 in this case). As a general rule MODSIM allows you to specify data in a variety of units but in some cases this would clutter the forms too much and when no alternatives for units are specified you should use the SI system. Internally MODSIM uses only SI units.

This is sufficient data to specify the ore characteristics for this exercise and you should complete the ore characteristics section of the system data form.

The feed material for this exercise is silica having a size distribution given in the table below.

Mesh size	850	600	425	300	212	150	106	75	53	38
microns										
% passing	95.7	88.3	77.0	63.3	49.6	37.4	27.3	19.6	13.8	10.2

The feed rate will be set at 50 tonnes/hr of solid and the feed slurry will contain 40% solids by mass.

You should now double click on the feed stream number in the box on the right hand side of the system data form and specify the properties of the feed.

The first thing to do is give the stream a name so that it can be identified easily later.

The next thing to do is to specify the size distribution. Note that the experimental data for the size distribution that is usually determined by screening will include less mesh sizes than the number that

Particle-size distribution and feed rate in feed streams

File

Specify a descriptive name for the stream. Must start with an alphabetic character

Mesh size	% Passing
Infinity	100.00
8.500E+2	95.70
6.000E+2	88.30
4.250E+2	77.00
3.000E+2	63.30
2.120E+2	49.60
1.500E+2	37.40
1.060E+2	27.30
7.500E+1	19.60
5.300E+1	13.80
3.800E+1	10.20

Clear

Stream number 1 Stream name Feed

Number of mesh sizes in your data 11

Units of size: ☒ micron ☐ mm ☐ cm ☐ m ☐ inch

☐ Use Rosin-Rammler distribution

Solids feed rate 1.3890E+1 Percent solids 40.00

Units of feed rate: ☒ kg/s ☐ Short tons/hr ☐ tonnes/hr ☐ Long tons/hr

Export size distribution

Import size distribution

Specify grade distributions

Specify distribution over S-classes

Cancel Accept

you previously specified for internal computations. You can specify just as many mesh sizes as are available and MODSIM will interpolate your data to match the internal size distributions. In the table above there are 11 size intervals (note not 10 as you might think if you count the columns in the table because the oversize on the top screen is not shown explicitly). So specify 11 mesh sizes in the form. Clear the default data and type in the data from the table. Remember to choose the units of size that you are using.

When this is complete, specify the feed rate and the percent solids in the feed in the appropriate boxes on the form.

Note that all other selections on the form are greyed out in this case because they are not required and MODSIM always tries to help the user to specify only the data that is necessary for the problem on hand. Accept the data when you are satisfied that it is what you want.

Finally you should name the overflow and underflow streams for future reference. To do this double click on the stream number in the "Internal and product streams" box. This will open the "Particle size distribution" form for the chosen stream. Specify the name for the stream and click Accept to return to the previous form. Do not attempt to add any additional data to these forms.

Save the job. This completes this exercise. Now go to exercise 2-2

Exercise 2-2 Specification of unit model parameters.

In this exercise you will use the job that you started in exercise 2-1. If this is not your current job, close the current job and open the job you previously saved in exercise 2-1.

Edit the unit model parameters.
(Open the main Edit menu).

For this exercise you should use the Plitt hydrocyclone model which you will find described in Modeling and Simulation of Mineral Processing Systems, Sections 4.5.4 - 4.5.7. The model is identified as model CYCL in MODSIM.

The screenshot shows a software window with the following elements:

- Job Name:** Exercise_2-1
- Unit type:** A table with two columns: 'Unit number' and 'Unit type'. The first row contains '1' and 'Hydrocyclone'.
- Models:** A list box containing three items: 'CYCL', 'CYCA', and 'NAGE'.
- Help:** A checkbox that is currently unchecked.

The flowsheet for this exercise includes only a single hydrocyclone so there is only one entry in the "Units" block of the "Select models and parameters for units" form. Click on "hydrocyclone" and the models that are available for a hydrocyclone will be listed in the "Models" list. Note MODSIM uses 4-character mnemonics to identify unit models. Check the "Help" box if you wish to see a brief description of the model before you specify the parameters. Double click on the model that you wish to use. If "Help" is checked you will see a description of the model including a description of the parameters that are required for the model. At this stage you can elect to accept the model or cancel your selection and try another model.

The forms that are provided to specify parameters for the unit models are unique and each model has its own form. Some are very simple and some are quite lengthy including more than 20 parameters for complex models such as those for autogenous mills.

For this exercise use the default values for all parameters except the cyclone diameter. Choose a cyclone of 60 cm diameter with standard geometry. Accept this data and click Accept on the "Select models and parameters for unit" form.

Save the job.

Open the main Run menu and click "Run simulation". If you do not get the "Simulation was completed successfully" and the "Data output completed successfully" messages you have specified some data badly. Review your system data and model parameters carefully.

MODSIM - Specify parameters for model CYCL for unit 1

File

Cyclone diameter: 60

Size specification by: ☒ Ratio ☐ Absolute

Vortex-spigot distance: 3

Inlet diameter: .2

Vortex finder diameter: .167

Spigot diameter: .116

Units of size: ☐ mm ☐ cm ☒ m ☐ inch

Head of feed slurry: 3 ☒ m ☐ feet

Slurry viscosity: .0012 ☒ Pa s ☐ c poise

Index for variation of d50 with density: .5

Factor for slurry density in separation zone (<1): 0

Number of cyclones in cluster: 1

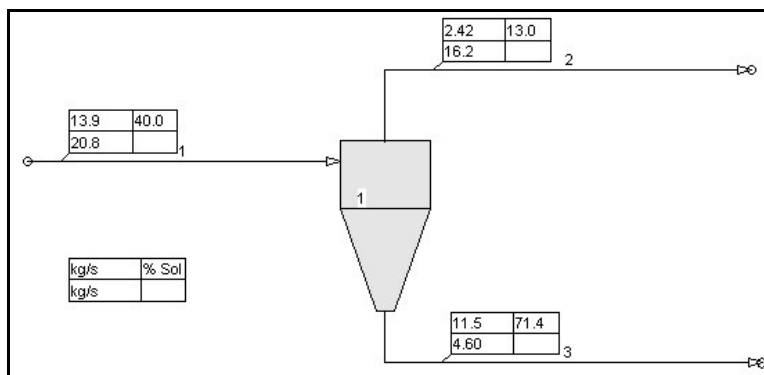
Data set: ☒ New ☐ Current data ☐ Default

Calibration factors:
 for d50: 1
 for the sharpness index: 1
 for the flow split: 1

Buttons: Cancel, Accept

It is unlikely that anything will go seriously wrong in this simple exercise so you should be able to see the results of the simulation now. This is a good time to save the job.

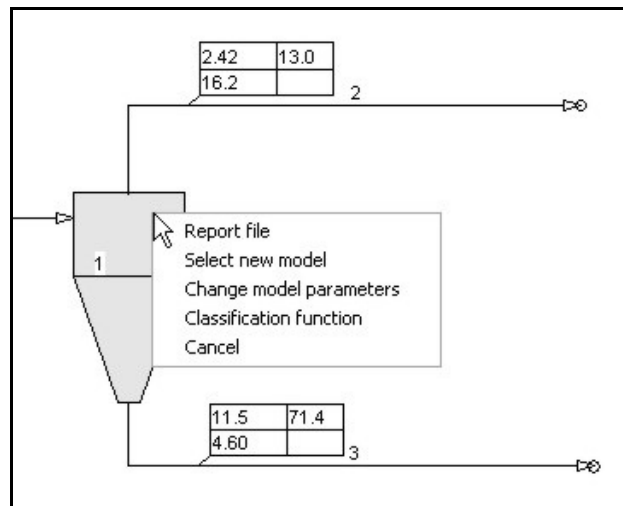
Open the main View menu and view the flowsheet. The flowsheet appears and you can get an immediate summary of the plant behavior from the data in the stream flyouts. Note in particular that the overflow has 13.0% solids and the underflow 71.4% solids. (If you do not see these values you have not specified some of the data as required and you ought to check all your data before proceeding. If you cannot find any error you should get job Exercise 2-1 from the Internet. The 71.4% solids in the underflow should catch your attention because a hydrocyclone will probably rope when discharging such a dense slurry. This will be discussed further later in this exercise.



The first thing to note are the units that are used to display the flowrates in the flyouts. The defaults (kg/s) should be showing in this exercise. In most problems you will want to choose more appropriate units. To do this you must edit the output format. (Without leaving this window, open the Edit menu and choose "Edit output format"). Select tonnes/hr in the "Units for

solid flowrate" block and cub-m/hr in the "Units for water flowrate" block. Uncheck the "Display latest output data" box and click "Accept". Note that the flyouts now display quantities using the units that you have selected.

The next type of data to look at is the report on the operation of the hydrocyclone. This is contained in the report file for the unit which is produced whenever the simulation is run. The easiest way to access the report file for the unit is to right click on the unit icon for the hydrocyclone and select "Report file" from the pop-up menu. A formatted file is displayed that is specific to the selected unit. Scroll through this file and note the kind of information that is displayed. In particular note the following items: the parameters used are shown for reference, the size distributions in the feed and product streams are tabulated and the calculated d_{50} size for the operating conditions ($85\text{ }\mu\text{m}$) is shown. Note that at the bottom of the file a warning is issued to the effect that the Mular-Judd criterion indicates that the cyclone will rope in operation. Furthermore the Concha analysis indicates that the air core diameter is 13% larger than the spigot diameter. This also indicates that roping can be expected.

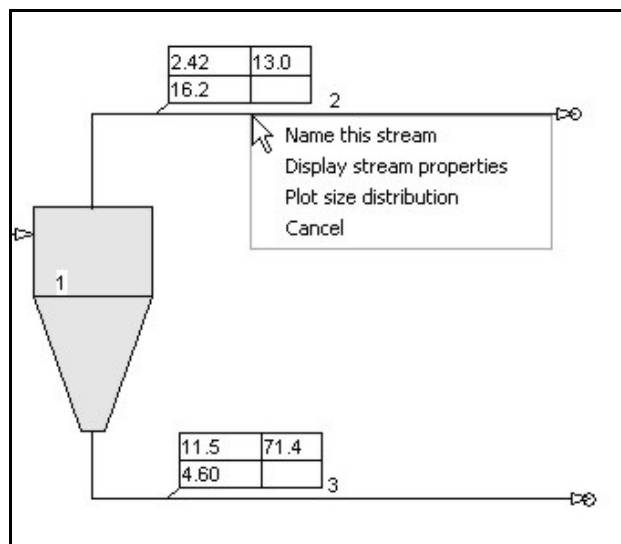


Before proceeding it is advisable to fix this potential problem. Recall that a 60 cm diameter cyclone was chosen with "standard" geometry. This requires the spigot diameter to be 11.6% the cyclone diameter. So now edit the unit parameters for the cyclone and set the spigot diameter to 15% of the cyclone diameter. The quickest and most convenient way to do this is to right click on the unit icon and select Change model parameters from the pop-up menu. This brings up the parameter specification form for the model and the spigot diameter can be changed.

RUN THE SIMULATION AGAIN by right clicking on any blank area of the flowsheet and left clicking the pop-up box or select Run simulation from the Run menu. Notice that the underflow is now 55.4% solids. Look at the report file for the hydrocyclone and note that d_{50} is now $71\text{ }\mu\text{m}$. At the bottom of the file you will note that the Mular-Judd criterion is still issuing a warning but the Concha formula indicates that operation is fine with an air core 92% of the spigot diameter. Experience has shown that the Mular-Judd criterion is too conservative and flags a warning too easily. Although it is always wise to take note of a warning like this, it is necessary to exercise

some judgement about it. (A well substantiated criterion for cyclone roping would be a valuable addition to the cyclone model so please let us know if you have any good operating data which could be used to develop a better criterion).

The next type of output data that are of interest are the particle size distribution graphs. These are accessed from the View menu by selecting View size distribution graphs. This will display the "Plot size distributions" window. Double click each of the three streams and these will be placed in the graph list. Click View graph to see the graphs. During the next step you will need to compare the overflow size distribution in the cyclone overflow stream with that obtained before the spigot diameter was changed. This requires the comparison of size distributions from different Modsim jobs. To do this, close the size distribution graph and save the data for the overflow stream by clicking the Save distribution for plotting button. This will allow you to navigate to a convenient folder and save the size distribution for the selected stream as a file. Remember to select the required stream in the stream list box before saving. Before leaving this job, make a note of the d_{80} , d_{50} and d_{20} sizes for the size distribution of the hydrocyclone overflow. This can be most easily done by right clicking anywhere on the overflow stream in the flowsheet. Select Stream properties from the pop-up menu.



This is a good time to change the job name to Exercise2-2 and save the job. Accept the flowsheet and select Change job name from the main Edit menu.

At this point you will probably want to see the difference in the overflow size distributions that resulted from the increase in spigot diameter. Run Exercise 2-1 again and look at the size distribution graphs. Import the saved size distribution from Exercise 2-2 so that this can be compared with the distribution from this rerun of Exercise 2-1. Look carefully and you will see that in spite of the smaller d_{50} size with the larger spigot the overflow size distribution is slightly coarser. Why? (You will need a sharp understanding of the behavior of hydrocyclones to answer this. A careful review of Section 4.4 in Modeling and Simulation of Mineral Processing Systems should provide the necessary information).

That completes exercise 2-2. You may wish to vary some of the other parameters in the cyclone model now and check the results. In fact you could really spend some time and put the Plitt model through its paces. You can also experiment with the other models CYCA and NAGE that are available for simulating hydrocyclone classifiers. You should now tackle exercise 2-3.

Exercise 2-3 Data for multicomponent ore.

This exercise will demonstrate the specification of data when the ore contains more than one mineral component. The problem studied in exercises 2-1 and 2-2 will be used but now the material to be processed is an iron ore consisting of 48% magnetite by mass in silica. To keep things simple assume that 80% of the silica is completely liberated and 75% of the magnetite is completely liberated. The remainder of the ore is unliberated and for simplicity assume that all incompletely liberated particles have the same composition or grade. (You will examine the correct way to get accurate representations of unliberated ore in Module 7).

Load the previous job from Exercise 2-2 if this is not your current job. Change the job name to Exercise 2-3 and save this job. Open the flowsheet editor to check that you have got the correct flowsheet and if so Accept the flowsheet to register the new job name on the flowsheet. Edit the system data. Change the number of minerals to 2, select specific gravity by mineral and add magnetite of specific gravity 5.2. Modsim allows you to specify the specific gravity of the particles by mineral or directly by particle type. In the former case the specific gravity of the composite particle will be calculated from the mineralogical composition of the particle using the mineral specific gravities. In the latter case Modsim uses the specific gravity of each particle type as specified by the user. Change the number of grade classes to 3, one for liberated silica, one for incompletely liberated particles and one for liberated magnetite. It is now necessary to specify the composition of the particles in each grade class. Click on Set up grade classes. This brings up a form that enables the grade of each particle type to be specified. Note that the default composition of the unliberated particles is 50% silica and 50% gangue. This must be changed in this example according to the following calculation.

Consider 100 mass units of ore:

$$\text{Units of liberated silica} = 52 \times 0.8 = 41.60$$

$$\text{Units of liberated magnetite} = 48 \times 0.75 = 36.00$$

$$\text{Units of unliberated ore} = 100 - 41.60 - 36.00 = 22.40$$

$$\text{Units of silica in unliberated portion} = 52 - 41.60 = 10.40$$

$$\text{Units of magnetite in unliberated portion} = 48 - 36.00 = 12.00$$

$$\text{Average silica grade in unliberated particles} = 10.40/22.40 = 0.4643$$

$$\text{Average magnetite in unliberated particles} = 12.00/22.40 = 0.5357.$$

Change the composition of grade-

Class	Composition	Sp. gr. of class
1	1.0000 0.0000	2.7
2	0.4643 0.5357	3.636599
3	0.0000 1.0000	5.2

Data set
☒ New
☐ Current
☐ Default

class 2 to 0.4643 0.5357. Note that you cannot change the specific gravity of the unliberated particles. MODSIM will calculate this for you because you previously specified that specific gravities are to be specified by mineral. The easiest way to get the calculation done is to select Default in the "Data set" box and then immediately reselect New which will show the new data with the specific gravity calculated. You can now accept this data.

It is now necessary to specify how the particles in the feed population are distributed over these three grade classes. To do this double click on the feed stream in the "Feed streams" box on the system data editing form. This will bring up the particle-size distribution form for the feed stream. The particle size distribution form opens with the "Specify grade distributions" button enabled. You will keep the size distribution used in the previous exercise so click on the "Specify grade distribution" button. This opens the form to specify the distribution of particles among the grade classes. From the calculation given above the mass fraction of the particles in G-class 1 is 0.4160, in G-class 2 (the unliberated particles) it is 0.2240 and in G-class 3 it is 0.3600. Clear the default data and enter these values.

Accept this data and the particle size data and the system data (Click three Accept buttons in succession). This will take you back to the main page. Take a

quick look at the unit parameters to make sure everything is OK and you are ready to run the simulation. Do it and then view the flowsheet to see the results.

Class	Mass fraction
1	0.416
2	0.224
3	0.360

Stream: 1 Feed

Size range: 0 - .002

Data set: ☒ New ☐ Current ☐ Default

We expect that the hydrocyclone should show an upgrading of the heavier mineral in the underflow and it is interesting to check this. A convenient way to do this is to show the Fe content of each stream in the vacant quadrant of each flyout. This is done using the output format facility so Edit the output format for the Edit menu.

Select your preferred units for the solid and water flowrates. Enter Fe into the first field under "Metals or elements". This will tell MODSIM that you want to keep track of iron in the flowsheet streams. Because MODSIM does all calculations using the individual minerals you must

Select quantities to display:

- ☒ Solids flowrate
- ☒ Water flowrate
- ☒ Recovery and grade of individual minerals
- ☒ % solids
- ☒ Yield of total solids
- ☒ Recovery and grade of individual metals

Units for solid flowrate:

- ☐ kg/s
- ☒ tonnes/hr
- ☐ ktonnes/month
- ☐ Mtonnes/year
- ☐ short tons/hr
- ☐ short tons/day

Units for water flowrate:

- ☐ kg/s
- ☒ tonnes/hr
- ☐ cub m/hr
- ☐ liters/min
- ☐ gal/min

Units for metal content:

- ☒ %
- ☐ g/t

Metals or elements:

	Fe			
Minerals				
Silica	0.0			
Magnetite	0.724			

Double click to select streams:

Selected streams:

Size: Accumulate

1 Feed
3 Underflow
2 Overflow

☐ Display size distributions of selected streams

☐ Accumulate data from selected streams

☐ Use format for coal data

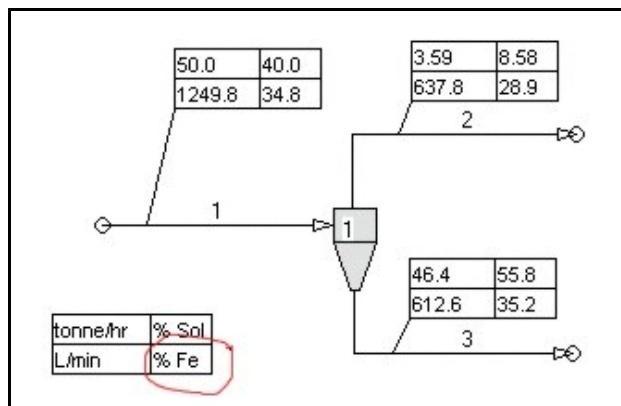
☒ Display latest output data

Cancel Accept

specify how much Fe is in each mineral that is listed - 0% in silica and 72.4% in magnetite (enter 0.724). To display the Fe content in each stream check the Recovery and grade of individual metals in the "Select quantities to display" frame. While your cursor is in the neighborhood, check the Recovery and grade of individual minerals because you will need this information shortly. Check the "Display latest output data" box and accept the form. The results from the simulation are displayed in a formatted file which you can examine. This file is most often used to insert simulation data into a report. Close this form and notice that the iron content of each stream is displayed in the 4th quadrant of the flyouts. Note the upgrading of the heavy mineral in the underflow.

Right click on the hydrocyclone icon and open its report file. Scroll down and check the d_{50} sizes for each G-class. Note that these decrease from G-class 1 to G-class 3. Why? Note also that d_{50} for silica is smaller now than in exercise 2-2. Why?

One final piece of information should be checked before you start investigating possible parameter changes. Recall that we never specified that the magnetite content of the feed should be 48%. MODSIM had to reconstruct this from the G-class information. To check this, View the simulation results from the View menu. Because the grade and recovery of the minerals was selected in the output format these are displayed in the table. The grade of magnetite in the feed is correctly calculated as 48%.



This concludes the exercises for module 2.

These jobs for exercises 2-1, 2-2 and 2-3 can be loaded into MODSIM from the Internet if you do not succeed in setting them up correctly yourself.

Module 3. Simulation of Crushing and Screening Circuits.**Objectives:**

1. Review the basic principles that are used to model the operation of crushers and screens.
2. Learn how to specify parameters for screen and crusher models.
3. Learn to understand the relationship between model parameters and the size distribution in the product streams.
4. Learn how to set up and simulate complete crushing plants.
5. Learn how to set up crushers and screens for best operation.

Schedule:

It is recommended that you tackle the segments in this module in the following order:

1. Vibrating screens
2. Crushers
3. Putting it all together - integrated crushing and screening plants. Exercise 3-5

Simulation the Performance of Vibrating Screens

We will examine two screen models that are based on the traditional method that is used to size screens for industrial applications. These models are based on a series of capacity factors that can be used to determine the tonnage of a particular material that can be handled by a screen of specified mesh size. The traditional method does not attempt to make a realistic calculation of the size distribution in the oversize and undersize products from the screen. However this traditional model is convenient for preliminary simulations particularly in the early stages of a project when not much data is available and the precise details of the screens that are to be used are not known.

A useful modification to the traditional model was made by Dr. V.K. Karra who used an extensive collection of industrial data to model the actual screening process in terms of the physical parameters that define the screen operation. This model is to be preferred when simulating a screening operation for which the physical properties of the screen are known such as the simulation of a screening operation in an existing plant. The Karra model has proved to be reliable when compared against industrial operating data and is the recommended screen model for real plant simulation.

These models are described in Sections 4.1 and 4.2 of Modeling and Simulation of Mineral Processing Systems. The more fundamental kinetic model for screening is also described in Section 4.3 but its use will not be covered in this course because there is as yet insufficient operating data to make good estimates of the specific transmission rates for typical industrial screens. It is hoped that these data will become available in the future.

The traditional model is demonstrated in exercise 3-1 and the Karra model is demonstrated in exercise 3-2.

Simulation of the Performance of Crushers

Several types of crushers are modeled in MODSIM. In this module we will consider three types: primary crushers (gyratory and jaw crushers), secondary crushers (standard cone crushers) and tertiary crushers (short head cone crushers). Models for crushers are described in Section 5.6 of Modeling and Simulation of Mineral Processing Systems.

Primary Crushers

The models used for primary crushers are elementary and follow a method that has been traditionally used by crusher manufacturers. The product size distribution is assumed to follow a standard pattern which is normalized with respect to a single reference size on the curve which is often d_{80} . The reference size is closely related to the open side setting of the crusher. The standard size distributions have been determined from many crushers operating in industry. In spite of the simplicity of this model, the results are adequate for many simulations since the product from these primary crushers is generally the feed to a further stage of crushing using standard or short head cone crushers. An improved form of the model developed by Csoke et al. is also available for primary crushers. The simulation of primary crushers is illustrated in exercise 3-3.

Standard and Short Head Cone Crushers

The models for both standard and short head cone crushers are based on the idea of internal breakage and classification developed by Whiten. This model allows calculations of realistic size distributions in the product given the size distribution in the feed and the closed side setting of the crusher. These models are described in Section 5.6 of Modeling and Simulation of Mineral Processing Systems. The model for the standard cone crusher is demonstrated in exercise 3-4 and for the short head crusher in exercise 3-5.

Exercise 3-1 Investigate the information generated by the traditional screen model in a typical operating environment.

Start a new job in MODSIM and draw a flowsheet containing only a single-deck screen.

Add a feed stream and product streams for oversize and undersize. Add flyouts to each stream.

Accept the flowsheet and edit the system data.

Use the following settings: 1 mineral sp. gr. 2.7, 25 size classes, maximum particle size 15 cms.

Specify the feed: 300 tonne/hr. at 95% solids with Rosin-Rammler distribution having $D_{63.2} = 4$ cm and $\lambda = 1.2$ Use 20 mesh sizes in the feed size distribution data.

Don't forget to name the two product streams for convenient reference later.

Edit the unit parameters. Use model SCRNM for the vibrating screen. Check the help box the first time you select the model to see a brief description of the model. Choose the following parameters: mesh

size 3 cm, transmission efficiency 90%, surface water on oversize 2%. Do not specify screen dimensions on the first run.

Accept the parameters and run the simulation.

View the flowsheet. Check the data in the stream flyouts which show the total tonnages in each stream.

Right click on the screen icon to see the report file for that unit. Examine the report file carefully to become familiar with the information that it provides. Note in particular that the basic capacity factor for this screen is 60.5 tonnes/hr. Note the calculated values of the six correction factors and that 5.76 sq.meters of screen area would be required for this duty.

View the size distribution graphs from the View menu.

Size distribution graphs and the report file for the screen are shown below for reference.

Modify some of the unit parameters and note how the performance data changes after each simulation run.

Save the job as Exercise 3-1.



Figure 21 Size distributions of the feed oversize and undersize using the traditional model.

Report file for the screen in Exercise 3-1

Unit number 1 MODSIM model name SCRN
Job name: Exercise 3.1 Traditional screen model

There is/are 1 screen/s inparallel

Tonnage to be processed 300.0 tonnes/hr.
Tonnage in undersize product 137.0 tonnes/hr.

Mesh size 30.00 mms.
Surface water on deck oversize 2.0 %

Factors are calculated for horizontal screen.

Basic capacity = 60.490 tons per hour per square meter

Particle size distribution in the feed.

Size mms	% passing
336.	100.000
238.	99.980
168.	99.632

119.	97.520
84.1	91.277
59.5	79.997
42.0	65.413
29.7	50.358
21.0	37.002
14.9	26.281
10.5	18.224
7.44	12.429
5.25	8.381
3.72	5.614
2.63	3.739
1.86	2.485
1.31	1.644
0.929	1.090
0.657	0.720
0.465	0.474
0.328	0.315
0.232	0.217
0.164	0.153
0.116	0.108
0.00	0.000

Percent oversize in feed 49.27%

Oversize factor is 1.164

Percent half size in feed 26.51%

Half-size factor is 0.730

Deck location factor 1.00

Specific gravity of material in feed 2.70

Bulk density of material in feed 1620. kgs/m**3

Bulk density factor is 1.013

Screen area required 5.764 sq meters

Particle size distribution in the overflow.

Size % passing

mms

336.	100.000
238.	99.962
168.	99.322
119.	95.435
84.1	83.944
59.5	63.183
42.0	36.340
29.7	9.269
21.0	6.810
14.9	4.837
10.5	3.354
7.44	2.288
5.25	1.542
3.72	1.033
2.63	0.688
1.86	0.457
1.31	0.303

0.929	0.201
0.657	0.133
0.465	0.087
0.328	0.058
0.232	0.040
0.164	0.028
0.116	0.020
0.00	0.000

Particle size distribution in the underflow

Size	% passing
mms	
336.	100.000
238.	100.000
168.	100.000
119.	100.000
84.1	100.000
59.5	100.000
42.0	100.000
29.7	99.240
21.0	72.919
14.9	51.793
10.5	35.915
7.44	24.493
5.25	16.516
3.72	11.063
2.63	7.368
1.86	4.897
1.31	3.240
0.929	2.148
0.657	1.419
0.465	0.934
0.328	0.620
0.232	0.427
0.164	0.302
0.116	0.213
0.00	0.000

Exercise 3-2 Karra model for vibrating screens.

Load the job from exercise 3-1. From the main Edit menu change the job name to Exercise 3-2.

Keep the same flowsheet consisting of a single screen and edit the system data to register the change in job name.

Edit the unit parameters. Note that the current model for the screen, SCRN, is shown at the top of the models list for the screen. Change the model by double clicking on SCR2 and check the help box to see a short description of the model. Set the screen mesh size to 3 cm and the screen length to 3.0 m and the screen width to 1.92 m. This gives a screen of area equal to that found necessary in exercise 3-1. Choose woven wire mesh with wire diameter 7.94 m. All other parameters can be left

at default values.

Run the simulation and view the flowsheet. Right click the screen icon to get the report file for the unit. Scroll through the report file to check the performance of the screen. Note the capacity factors and note that the area utilization factor is 1.11 This parameter has been found to be useful in assessing the results of a simulation of screening behavior using the Karra model. It is calculated as the ratio between the actual amount of undersize that is transmitted by the screen to the amount of undersize that the screen is capable of transmitting as calculated using the Karra model. Thus it gives a measure of the loading on the screen relative to its theoretical capacity. An area utilization factor less than unity indicates that the screen is under loaded while a value greater than unity indicates that the screen is over loaded. Since the screen chosen had the area suggested by the traditional model in Exercise 3-1, the Karra model indicates that the traditional model under estimates the required area by about 11% with the medium heavy woven wire mesh that we have chosen. Note that the report file also indicates that the screen performance may be improved by installing water sprays

Look at the size distribution in the three streams from the view menu. The report file is shown below for reference.

You can now investigate a number of what-if scenarios. An instructive change is the addition of water sprays to the screen. This can increase the capacity of the screen. To do this the flowsheet must be edited and a water addition stream

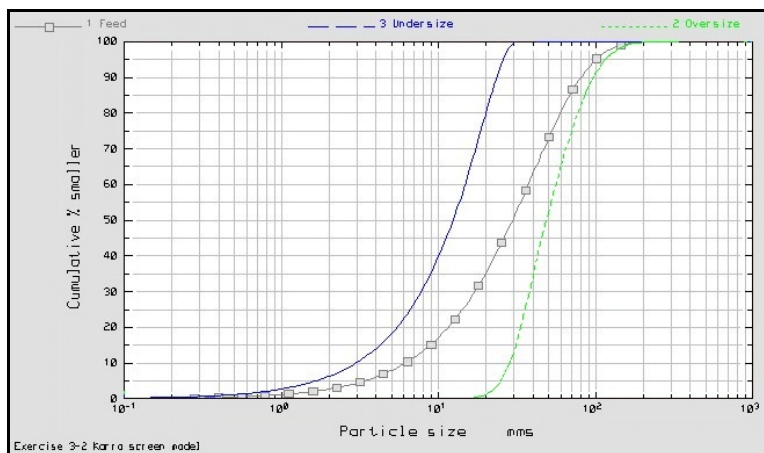


Figure 22 Simulated size distributions using the Karra model.

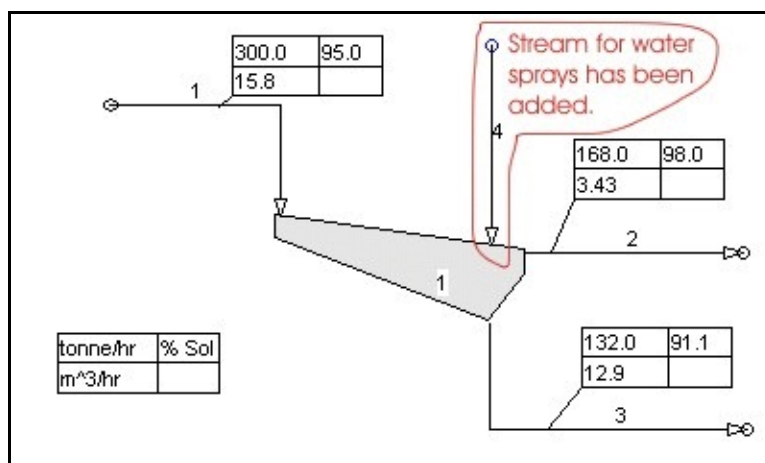


Figure 23 Water sprays can be added to the screen.

inserted to supply the spray water to the screen.

Accept the flowsheet and edit the System Data to specify the required spray water rate. Double click on the stream number that identifies the newly added stream to open the water stream form as shown below. Give the stream a name for subsequent reference and specify a water rate of 0.5 cubic meters per hour.

Figure 24 Form to specify water addition rate.

Accept this data and run the simulation again. View the flowsheet and note any differences in the results.

In particular check the value of the area utilization factor in the report file for the screen. You will notice that the screen is now matched exactly to the duty with an area utilization factor close to 1.0. There are many other What-If scenarios that would be worth investigating with this simple flowsheet.

You can now investigate a number of what-if scenarios. For example check how the area utilization factor varies as the wire diameter changes. Standard wire diameters for this mesh size are: heavy duty 9.53 mm, medium heavy 7.94 mm, medium 6.35 mm and medium light 5.72 mm.

You could also investigate how the area utilization factor varies as the feed rate changes.

You could also substitute a double check screen and simulate its performance.

Report file for exercise 3-2

```
Unit number      1          MODSIM model name SCR2
Job name: Exercise 3.1 Traditional screen model
```

```
Tonnage to be processed      :   300.    tons/hour.
Screen length is             :    3.84 m
Screen width is              :    1.50 m
Mesh aperture is             :    3.00 cms
Wire diameter is             :    0.85 cms
Screen is inclined at 0.00 degrees to the horizontal
```

PARTICLE SIZE DISTRIBUTION IN THE FEED.

Size cms	% passing
33.6	100.000
23.8	99.980
16.8	99.632
11.9	97.520
8.41	91.277
5.95	79.997
4.20	65.413
2.97	50.358
2.10	37.002
1.49	26.281
1.05	18.224
0.744	12.429
0.525	8.381
0.372	5.614
0.263	3.739
0.186	2.485
0.131	1.644
0.929E-01	1.090
0.657E-01	0.720
0.465E-01	0.474
0.328E-01	0.315
0.232E-01	0.217
0.164E-01	0.153
0.116E-01	0.108
0.00	0.000

Calculated throughfall aperture : 3.00 cms
 Percent oversize in feed : 49.271 %
 Percent half size in feed : 26.515 %
 Near size in feed : 20.930 %
 D50 for separation : 2.54 cms
 Percent moisture in feed : 5.00 %
 Bulk density of material in feed : 1603.00 kgs/m**3

Basic capacity factor A : 22.301 tph/m*m
 Oversize factor B : 1.009
 Half-size factor C : 1.018
 Deck location factor D : 1.000
 Wet screening factor E : 1.000
 Material weight factor F : 1.001
 Near-size factor Gc : 0.865

Tonnage in underflow : 129.85 tph
 Total capacity factor (B*C*D*E*F*Gc) : 88.87 %
 Simulated efficiency : 85.33 %

Area utilisation factor : 1.14

WARNING: This screen is overloaded. A larger screen should be considered.

Calculated rate of travel : 0.24 m/s

Bed depth at feed end : 14.18 cms
 Bed depth at discharge end : 8.04 cms

PARTICLE SIZE DISTRIBUTION IN THE OVERFLOW.

Size cms	% passing
33.6	100.000
23.8	99.964
16.8	99.351
11.9	95.627
8.41	84.620
5.95	64.731
4.20	39.017
2.97	12.694
2.10	1.680
1.49	0.171
1.05	0.016
0.744	0.001
0.525	0.000
0.372	0.000
0.263	0.000
0.186	0.000
0.131	0.000
0.929E-01	0.000
0.657E-01	0.000
0.465E-01	0.000
0.328E-01	0.000
0.232E-01	0.000
0.164E-01	0.000
0.116E-01	0.000
0.00	0.000

PARTICLE SIZE DISTRIBUTION IN THE UNDERFLOW

Size cms	% passing
33.6	100.000
23.8	100.000
16.8	100.000
11.9	100.000
8.41	100.000
5.95	100.000
4.20	100.000
2.97	99.710
2.10	83.284
1.49	60.494
1.05	42.083
0.744	28.712
0.525	19.362
0.372	12.969
0.263	8.638
0.186	5.741
0.131	3.798
0.929E-01	2.518

0.657E-01	1.664
0.465E-01	1.095
0.328E-01	0.727
0.232E-01	0.500
0.164E-01	0.354
0.116E-01	0.250
0.00	0.000

NOTES:

The screen was shown as operating without water sprays
Using water sprays could increase the capacity of the screen.

Surface water retained on the oversize 2.00 %

Exercise 3-3. Gyratory crusher

The next three exercises step you through the method to develop a crusher circuit to produce 1200 tonnes/hr of aggregate having 80% passing 10 mm.

Start a new MODSIM job.

Select a single gyratory crusher for the flowsheet and add a feed stream and a product stream.

Edit the system data and select one mineral of specific gravity 2.7.

Set the largest size to 1.2 m with 25 size classes.

Double click on the number of the feed stream to specify feed stream data. Name the feed stream, show 20 size classes, and use Rosin-Rammler distribution with $D_{63.2} = 40$ cm, $\lambda = 1.4$. Remember to click the clear button to generate the R-R distribution.

Name the product stream before accepting the system data.

Edit the unit parameter and select model GYRA for the gyratory crusher. Check the help box if you want to see a brief description of the model.

Select the following parameters:
Open side setting OSS = 15 cm, WI = 12 kWh/tonne. Average ore type.
Edit the output format and select tonne/hr for solid flowrate.

Run the simulation.

View the particle size distributions in the feed and product streams.

View the flowsheet. Right click on the crusher icon to view its report

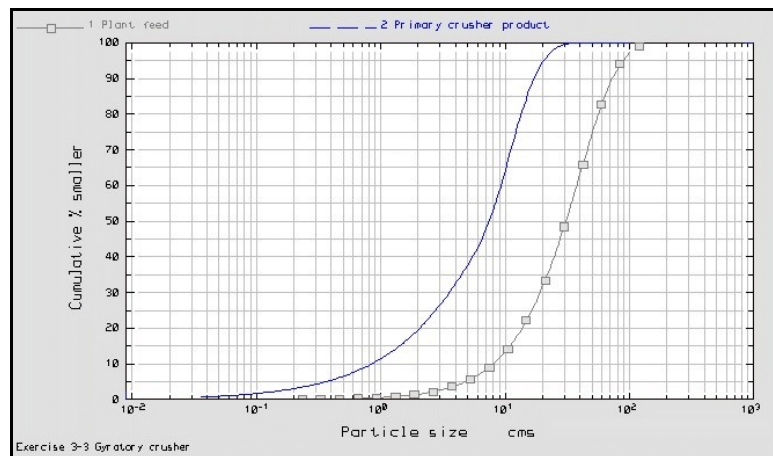


Figure 25 Size distributions in the feed and product streams

file. Note that the output includes a table of size distributions for the crusher feed and the crusher product. An estimate of the power required is also calculated using the crushing work index and the d_{80} sizes of feed and product. You should investigate how the size distribution of the crusher product varies as the open side setting is varied.

Report file for Gyratory Crusher in exercise 3-3

Unit number 1 MODSIM model name GYRA
Job name: Exercise 3-3 Gyratory crusher

Tonnage to be processed 1500. tons per hour

Open side discharge setting 15.00 cms

Size distribution in feed

Size cms	% passing
101.	97.407
71.3	89.447
50.5	74.941
35.7	57.340
25.2	40.812
17.8	27.594
12.6	18.028
8.92	11.518
6.30	7.257
4.46	4.529
3.15	2.812
2.23	1.739
1.58	1.076
1.12	0.668
0.788	0.412
0.558	0.256
0.394	0.157
0.279	0.093
0.197	0.059
0.139	0.042
0.985E-01	0.030
0.697E-01	0.021
0.493E-01	0.015
0.348E-01	0.010
0.00	0.000

Top size in feed is approximately 119.09 cms

Dimension A of crusher must be at least 148.86 cms

Minimum head diameter required to handle the tonnage.....

"Fall through" material in feed 22.38 percent.

Scalping of the feed is recommended.

Size distribution in product

Size cms	% passing
101.	100.000
71.3	100.000
50.5	100.000
35.7	99.905
25.2	98.403
17.8	91.456
12.6	76.851
8.92	58.107
6.30	43.946
4.46	35.024
3.15	27.470
2.23	21.276
1.58	16.321
1.12	12.428
0.788	9.412
0.558	7.098
0.394	5.337
0.279	4.003
0.197	2.996
0.139	2.240
0.985E-01	1.674
0.697E-01	1.249
0.493E-01	0.932
0.348E-01	0.695
0.00	0.000

Impact work index of the material is 12.00 kWhr/ton

80% passing size in feed is 561.96 mms.

80% passing size in product is 134.43 mms.

Estimated power requirements 251. kW

Power requirement to allow for 35% surge capacity 338.6 KW.

Crusher selected.....

Crusher capacity from manufacturer's tables.....tons per hour.

Number of crushers required.....

Exercise 3-4. Two-stage crushing

Edit the flowsheet from exercise 3-3 and add a second-stage cone crusher to handle the product from the gyratory crusher that was used in exercise 3-3. Since cone crusher capacity is less than the gyratory you must either use more than one secondary crusher in parallel or, preferably, screen the primary crusher product and send only the oversize to the second-stage crusher.

Change the jobname to Exercise 3-4 and edit the system data to name the additional product stream. Edit the unit parameters for the additional crusher, use the model CRSH, and set the the closed-side setting CSS to 35 mm. Leave other parameters at their default values.

Edit the parameters for the screen. Use model SCRNM with 45 mm mesh.

Run the simulation and view the flowsheet. Right click on the cone crusher icon to view the report file for this unit. Check the crusher capacity in the crusher capacity tables 5.4 to 5.7 in Modeling and Simulation of Mineral Processing Systems . One 7-ft standard crusher would be required to handle the tonnage.

Right click on the streams to find the d_{80} size in the plant feed, the primary crusher product and the secondary crusher product. Calculate the reduction ration in each stage by calculating the ratio of the d_{80} sizes in the stage feed and product.

Plot the size distributions in the plant feed, the primary crusher product, the secondary crusher product and the final product which will constitute the feed to the next stage of crushing which will be examined in Exercise 3-5
Save this job as Exercise 3-4.

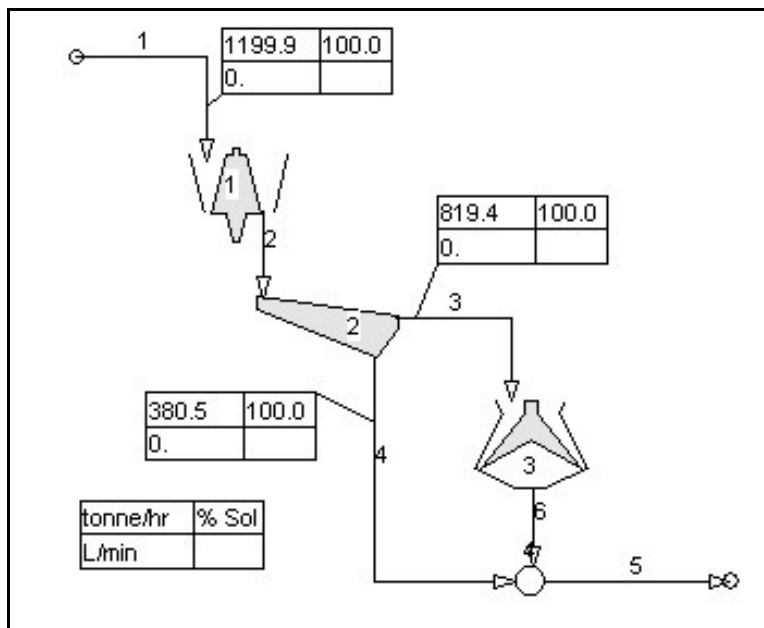


Figure 26 Flowsheet with 2 crushing stages.

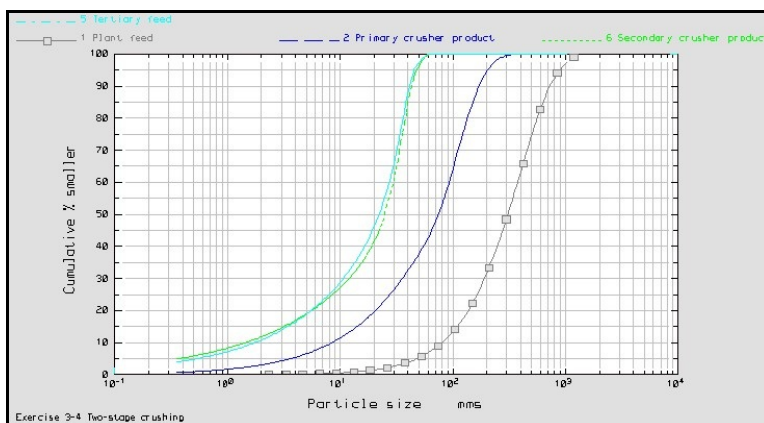


Figure 27Size distributions in 2-stage crushing circuit.

Report file for secondary crusher in
Exercise 3-4

Unit number 3 MODSIM model name CRSH

Job name: Exercise 3-4 Two-stage crushing

PARAMETERS:

Fines fraction 0.2000

Classification function parameters 0.653 1.600
 Tonnage to be processed 819.3 tons per hour
 Closed side setting 35.00 mms

Size distribution in feed

Size mms	% passing
0.101E+04	100.000
713.	100.000
505.	100.000
357.	99.861
252.	97.661
178.	87.487
126.	66.095
89.2	38.641
63.0	17.901
44.6	5.130
31.5	4.023
22.3	3.116
15.8	2.390
11.2	1.820
7.88	1.378
5.58	1.040
3.94	0.782
2.79	0.586
1.97	0.439
1.39	0.328
0.985	0.245
0.697	0.183
0.493	0.136
0.348	0.102
0.00	0.000

Top size in feed is approximately 286. mms
 Open side feed opening must be at least 358. mms

Size distribution in product

Size mms	% passing
0.101E+04	100.000
713.	100.000
505.	100.000
357.	100.000
252.	100.000
178.	100.000
126.	100.000
89.2	100.000
63.0	100.000
44.6	92.028
31.5	64.666
22.3	45.300
15.8	35.176
11.2	28.824
7.88	23.967

5.58	20.009
3.94	16.729
2.79	13.995
1.97	11.713
1.39	9.809
0.985	8.217
0.697	6.887
0.493	5.773
0.348	4.842
0.00	0.000

Select a crusher size and cavity type to suit the closed side setting and feed opening requirements.

Crusher selected is.....

Capacity in tons per hour?

Number of crushers required?....

Impact work index of the material is 12.00 kWhr/ton

80% passing size in feed is 155.74 mms.

80% passing size in product is 37.39 mms.

Estimated power requirements 259. kW

Exercise 3-5. Three-stage crusher circuit

In this exercise you will add a third stage of crushing to the flowsheet to reduce the size to meet the requirement that the product should have a d_{80} size of 10 mm. The third stage will be operated in closed circuit, and this will introduce you to the simulation of plants with recycle.

Open the job that you saved as exercise 3-4. Change the job name to Exercise 3-5.

Add a screen and a cone

crusher and connect the final product as the feed to the screen. (You can draw the screen icon then move it so that it just touches the arrowhead of the product stream. MODSIM will recognize the connection when you refresh the flowsheet.) Direct the screen oversize to the crusher. At this stage it is not advisable to close the circuit, and you should run the simulator in open circuit at least once

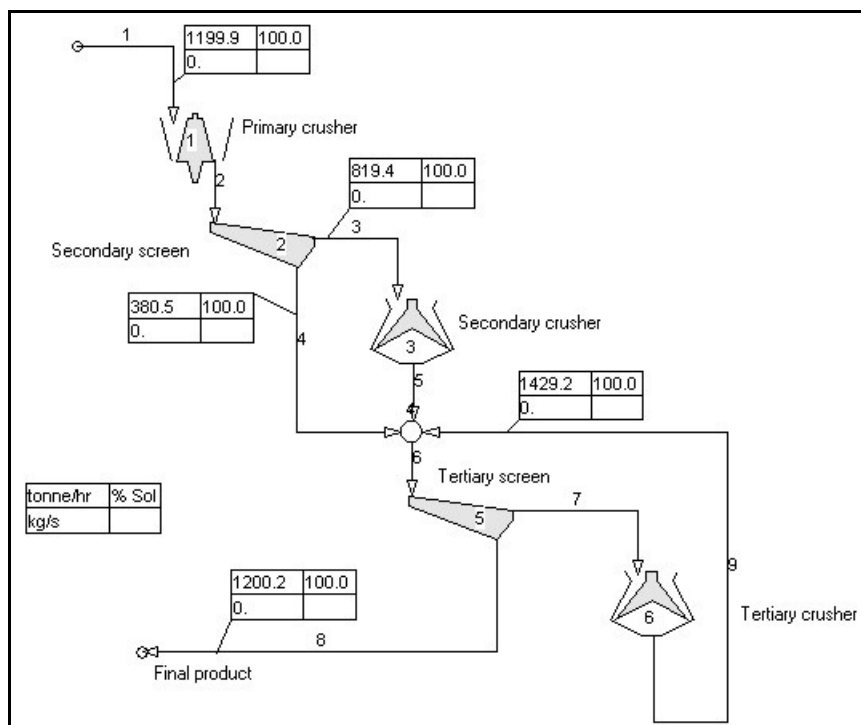


Figure 28 Flowsheet for 3-stage crusher circuit.

to make sure that all parameters are properly set up.

Use model SCRIN for the screen with a mesh size of 13 mm. Choose the short head model SHHD for the crusher and set the closed side setting CSS to 13 mm.

Run the simulation and check the report files for the two units that have been added. You should also check the size distributions in the products from the screen and the mill. When you are satisfied that the units are functioning properly, direct the crusher product back to the screen. Remember that the screen icon can accept only one feed stream so that a mixer or sump must be inserted in the flowsheet ahead of the screen to receive the secondary screen undersize, the secondary crusher product, and the tertiary crusher product.

Run the closed-circuit simulation and MODSIM should converge easily. If you are using a slow computer (<100 MHz), you should notice that the calculation does take a little longer because the recirculating load is calculated by iteration. In the unlikely event of some difficulty with the iterative calculation, MODSIM will

give you some diagnostic information to pinpoint the trouble. By far the most common cause of difficulty with iterative calculations is caused by inappropriate choice of model parameters. Check these thoroughly and you should find your error. If you cannot diagnose the problem, download the exercise job from the

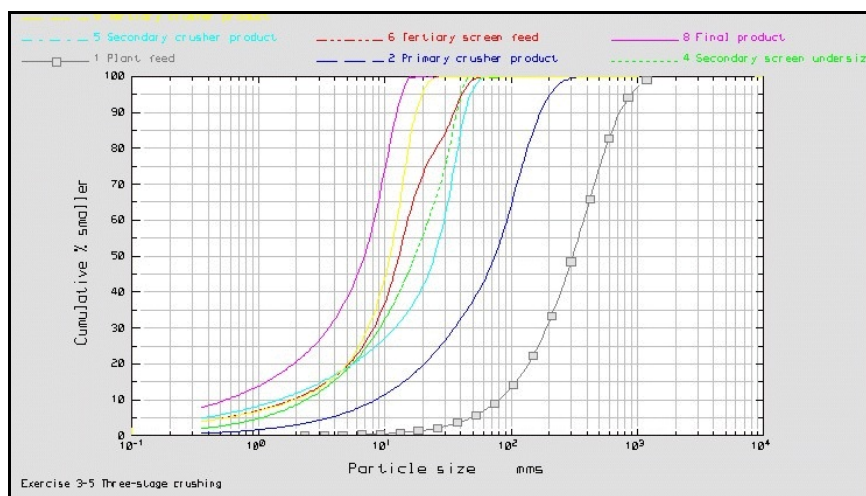


Figure 29 Size distributions in 3-stage crusher circuit.

course FTP site and investigate any differences that you can find. If you cannot sort out a problem, pack the job and send it to your instructor. It is most important that you become comfortable with simulating flowsheets that have recycle streams because this is a common feature of many industrial plants.

Plot that size distributions in all significant streams in the plant. Get the d_{80} sizes for the plant feed and the final product and calculate the overall size reduction ratio for the plant.

When you are comfortable with the simulation, you could try any number of what-if scenarios. Try all or some of the following:

1. Vary the mesh size of the tertiary screen over the range 8 mm to 15 mm and observe how the recirculating load and the size distribution of the final product change.

2. Vary the mesh size and the CSS in the tertiary circuit and seek out an optimal combination that will minimize the size and number of crushers required. Also keep track of the area requirements for the screen.

3. Reverse the order of the tertiary screen and the tertiary crusher so that the entire product from the secondary circuit passes through the crusher which is then screened to close the circuit.

```
Unit number      6          MODSIM model name SHHD
Job name: Exercise 3-5 Three-stage crushing
PARAMETERS:
Fines fraction  0.2000
Breakage function  $K(x/y)^{0.518} + (1-K)(x/y)^{2.475}$ 
Classification function parameters  0.653  2.000 n=2
```

Tonnage to be processed 1429. tons per hour

Closed side setting 13.00 mms

Size distribution in feed

Size mms	% passing
0.101E+04	100.000
713.	100.000
505.	100.000
357.	100.000
252.	100.000
178.	100.000
126.	100.000
89.2	100.000
63.0	100.000
44.6	95.259
31.5	73.863
22.3	56.550
15.8	30.429
11.2	7.655
7.88	5.326
5.58	3.882
3.94	2.981
2.79	2.363
1.97	1.908
1.39	1.558
0.985	1.279
0.697	1.055
0.493	0.872
0.348	0.722
0.00	0.000

Top size in feed is approximately 48.6 mms

Open side feed opening must be at least 60.8 mms

Size distribution in product

Size mms	% passing
0.101E+04	100.000
713.	100.000
505.	100.000
357.	100.000
252.	100.000
178.	100.000
126.	100.000
89.2	100.000
63.0	100.000
44.6	100.000
31.5	100.000
22.3	98.470
15.8	81.899
11.2	50.630
7.88	32.409
5.58	21.984
3.94	16.187
2.79	12.584
1.97	10.103
1.39	8.259
0.985	6.817
0.697	5.656
0.493	4.706
0.348	3.922
0.00	0.000

Select a crusher size and cavity type to suit the closed side setting and feed opening requirements.

Crusher selected is.....

Capacity in tons per hour?

Number of crushers required?....

Impact work index of the material is 12.00 kWhr/ton

80% passing size in feed is 34.04 mms.

80% passing size in product is 15.39 mms.

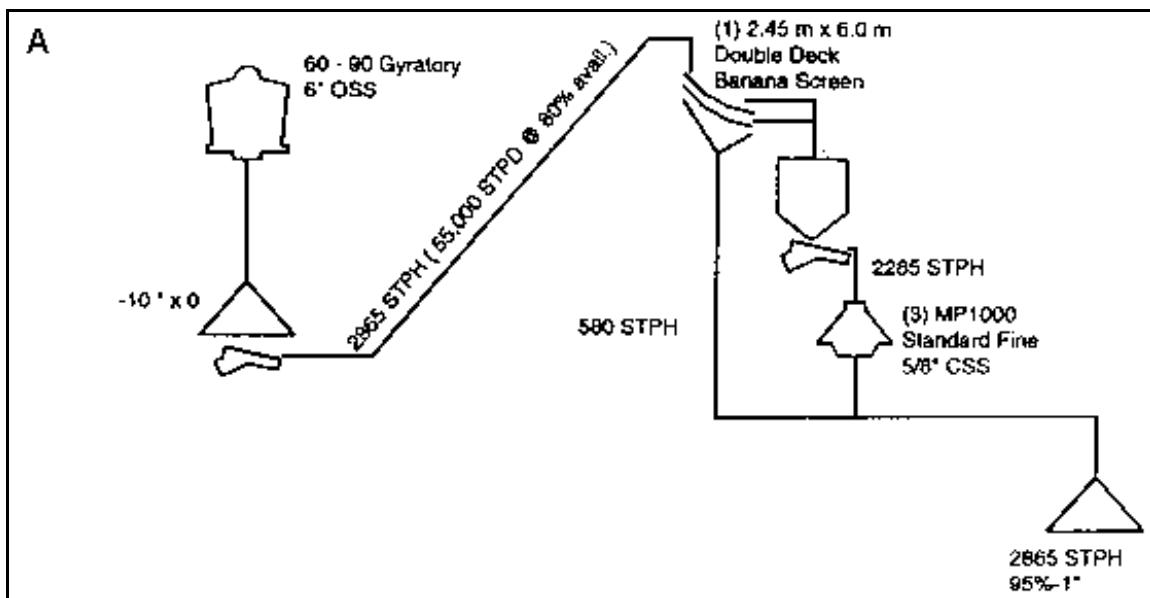
Estimated power requirements 453. kW

Assignment 3-1 Crusher circuits

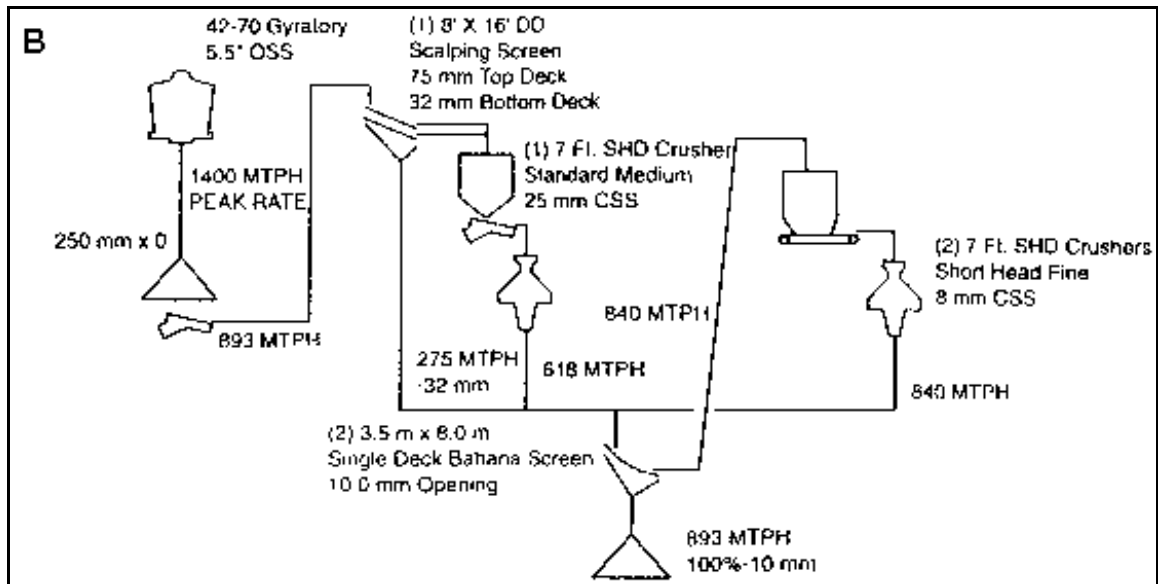
The flowsheets shown below are taken from a recent Nordberg product brochure. Four different crusher flowsheets are shown and these make nice subjects for simulation. Simulate the performance of each of these circuits. Report the size distribution in the final product from each flowsheet and use the standard equipment capacities for crushers from the textbook to assess whether the crusher sizes that are shown are appropriate. Calculate the energy required for the crushing operations if the crusher work index of the material is 12.5 kWhr/tonne in each case. Make what ever assumptions you consider appropriate if essential data items are missing. A Nordberg C160B jaw crusher has capacity close to the generic 48x60 jaw crusher in the standard capacity table. Assess the state of loading on each screen and thus determine whether the screen dimensions are appropriate for the duty.

Feed size distributions in each case may be assumed to be Rosin-Rammler with the following parameters.

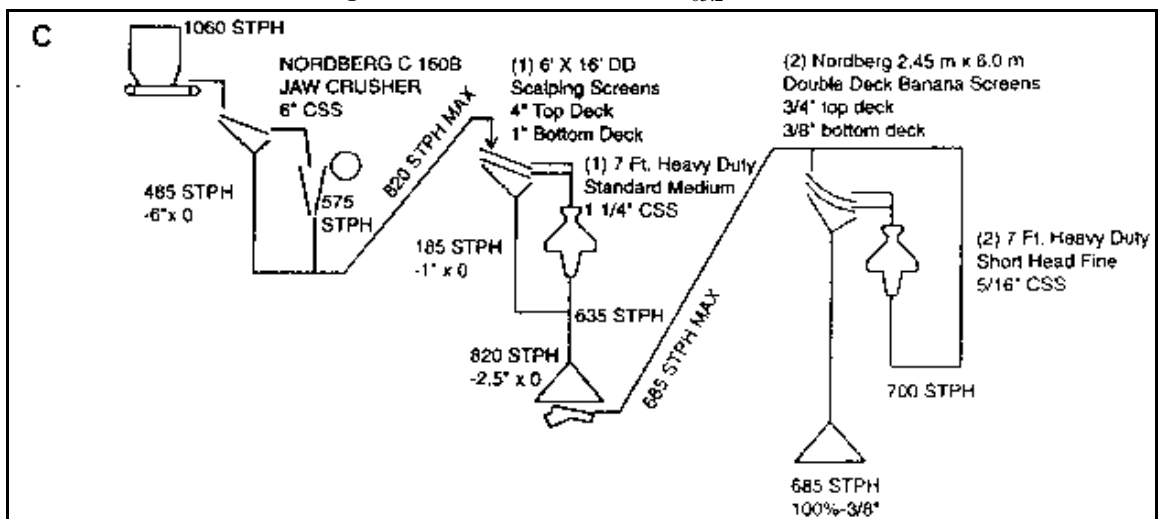
Flowsheet A: Rosin-Rammler parameters for the feed: $D_{63.2} = 20$ inches, $\lambda = 1.26$



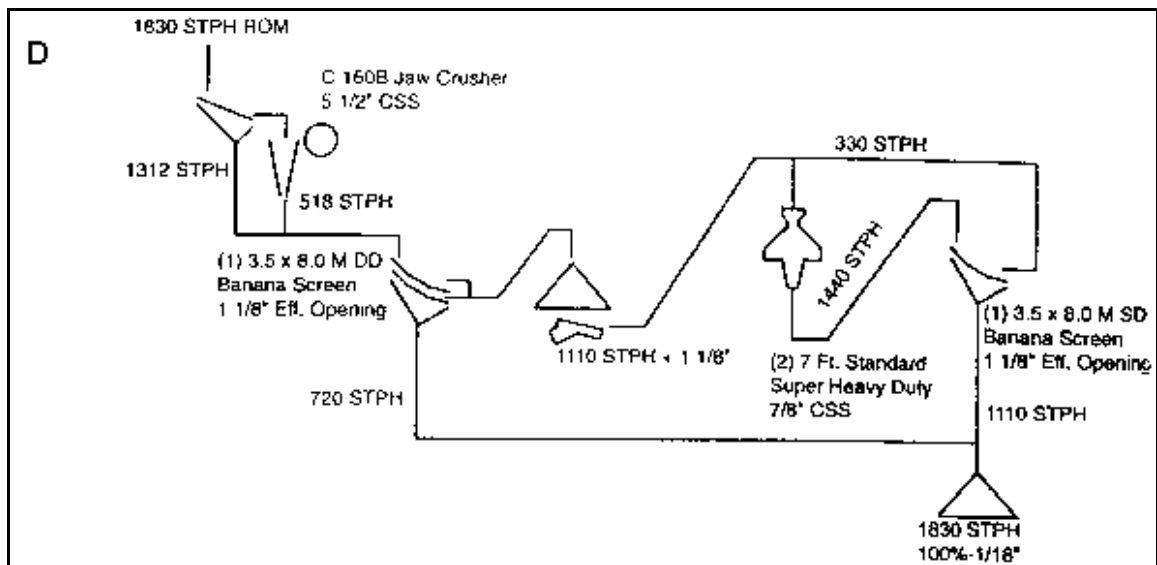
Flowsheet B: Rosin-Rammler parameters for the feed: $D_{63.2} = 21$ inches, $\lambda = 1.16$



Flowsheet C: Rosin-Rammler parameters for the feed: $D_{63.2} = 7.6$ inches, $\lambda = 2.5$



Flowsheet D: Rosin-Rammler parameters for the feed: $D_{63.2} = 3.6$ inches, $\lambda = 1.53$



Assignment 3-2 Aggregate preparation plant

A crushing and screening plant used for the production of building aggregate is shown in Figure 1. The equipment schedule is given in Table 1. The plant produces three products.

The plant processes 400 tonnes/hr of feed and the size distribution in the feed is given in Table 2.

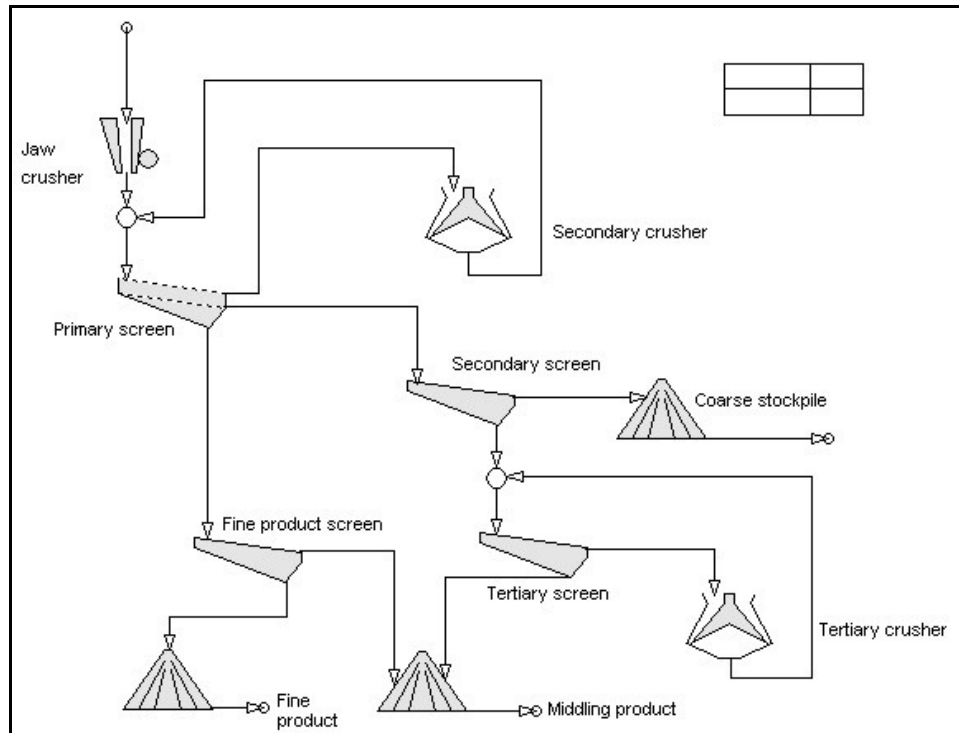


Figure 34 *Flowsheet of an aggregate plant*

Simulate the performance of this plant and investigate how the relative production rates of the three products, and their size distributions, vary as the closed side setting of the secondary crusher and the mesh size of the lower deck of the primary screen vary. You should be aware of any possible overload conditions that may occur as plant conditions are varied.

To display the results you should plot the production rates against the two control variables. To display the variations in the size distributions, plot the D_{80} , D_{50} and the D_{20} sizes against the mesh size of the screen.

Table 1 Equipment schedule for aggregate plant

Unit	Type	Settings
Primary crusher	Jaw crusher	2 × 630mm × 1000mm OSS 90mm
Primary screen	Double-deck inclined wire-mesh screen	1.83m × 3.66m 15° inclination Top deck: 90mm mesh opening 16mm wire Lower deck: 6mm mesh opening 3.4mm wire
Secondary crusher	Standard Symons cone	1.78m coarse cavity CSS 63mm
Secondary screen	Single-deck inclined wire-mesh screen	1.83m × 4.27m 15° inclination 25mm mesh 7.94mm wire
Tertiary screen	Single-deck inclined wire-mesh screen	1.52m × 4.27m 15° inclination 10mm mesh 4.1mm wire
Tertiary crusher	Short head cone	1.22m medium cavity CSS 6mm
Final product screen	Single-deck inclined wire-mesh screen	1.22m × 3.66m 15° inclination 1.5mm mesh 1.19mm wire

Table 2 Particle size distribution in the feed

Size mm	% passing
300	96
150	74
75	52
37.5	34
19.0	20
13.2	12
9.5	7
6.7	4
4.75	3
3.35	2
2.36	1

